

Beyond Non-Maximum Suppression

A Comparative Latency and Accuracy Analysis of YOLO26 vs RT-DETR
on the Symile-MIMIC Multimodal Dataset

Project Progress Review

Team:

- Aditya Kotkar
- Rishi Waghmare
- Jidnyesh Suryawanshi
- Shivam Hippalgave

Guide:

Mrs. Diptee Chikmurge
School of Computer Engineering,
MIT Academy of Engineering

Introduction & Problem Statement

- ▶ **Traditional Dependency:** Object detection in medical imaging relies heavily on Non-Maximum Suppression (NMS) to eliminate redundant predictions.
- ▶ **Latency Bottleneck:** NMS introduces additional computational latency and heavily relies on heuristic post-processing parameters.
- ▶ **Clinical Risks:** Small, dense, or overlapping lesions and abnormalities can be mistakenly suppressed and discarded by NMS logic.
- ▶ **The Objective:** Safety-critical healthcare domains require models that guarantee high accuracy, minimal latency, and strict structural reliability without heuristics.

The Proposed Solution

Shift to end-to-end architectures that directly learn one-to-one prediction mappings, entirely eliminating the need for heuristic NMS post-processing.

YOLO26

CNN-Based Paradigm

Represents the cutting edge of convolutional approaches. It utilizes optimized loss functions and efficient feature extraction optimized for high-throughput.

Strengths: Low-latency, real-time inference, and deployment efficiency on edge devices.

Core Tech: Progressive Loss Balancing (ProgLoss) and Small-Target-Aware Label Assignment (STAL).

RT-DETR

Transformer-Based Paradigm

Represents the shift towards set-prediction problems. It leverages hybrid encoders to combine convolutional speed with transformer context.

Strengths: Global attention mechanisms, improved contextual understanding, and robust small-target detection.

Core Tech: Attention mechanisms and Bipartite Matching algorithms.

System Architecture & Methodology



Dataset & Prep

Utilizing the Symile-MIMIC multimodal chest X-ray dataset. Images are heavily preprocessed with pixel normalization, contrast enhancement, and augmentations to handle subtle pathological features.



Pseudo-BBox Gen

A DenseNet121 CheXpert model generates Grad-CAM heatmaps for 5 multi-label pathologies. These are dynamically converted into pseudo-bounding boxes using thresholding and morphological operations.



Dual-Architecture

Simultaneous evaluation. YOLO26 uses localized CNN receptive fields, while RT-DETR processes tokens through a hybrid encoder and decoder to establish one-to-one mappings.

Mathematical Model

Set-Based Matching & Loss

Eliminates NMS by establishing a strict one-to-one assignment using bipartite Hungarian matching.

$$\sigma^* = \arg \min_{\sigma \in \mathcal{S}_N} \sum_{i=1}^M \mathcal{L}_{match}(y_i, \hat{y}_{\sigma(i)})$$

where:

- σ is a permutation function
- $\mathcal{S}_N$ is the set of all permutations
- $\mathcal{L}_{match}$ is the matching cost function

Total loss combines classification and regression objectives:

$$\mathcal{L} = \lambda_{cls} \mathcal{L}_{cls} + \lambda_{box} \mathcal{L}_{box}$$

Bounding Box & Complexity

Regression loss integrates both the L1 spatial distance and the Intersection over Union (IoU) metric.

Bounding Box Loss:

$$\mathcal{L}_{box} = \|b_i - \hat{b}_i\|_1 + (1 - IoU(b_i, \hat{b}_i))$$

Code Implementation Details



Pipeline Infrastructure: End-to-end Python script built efficiently using PyTorch, Ultralytics, and TorchXRayVision libraries for deep learning.



Automated Dataset Processing: Developed a fault-tolerant script to extract massive MIMIC .npy arrays, seamlessly converting them into an 80/20 train/validation split in strictly formatted YOLO layouts.

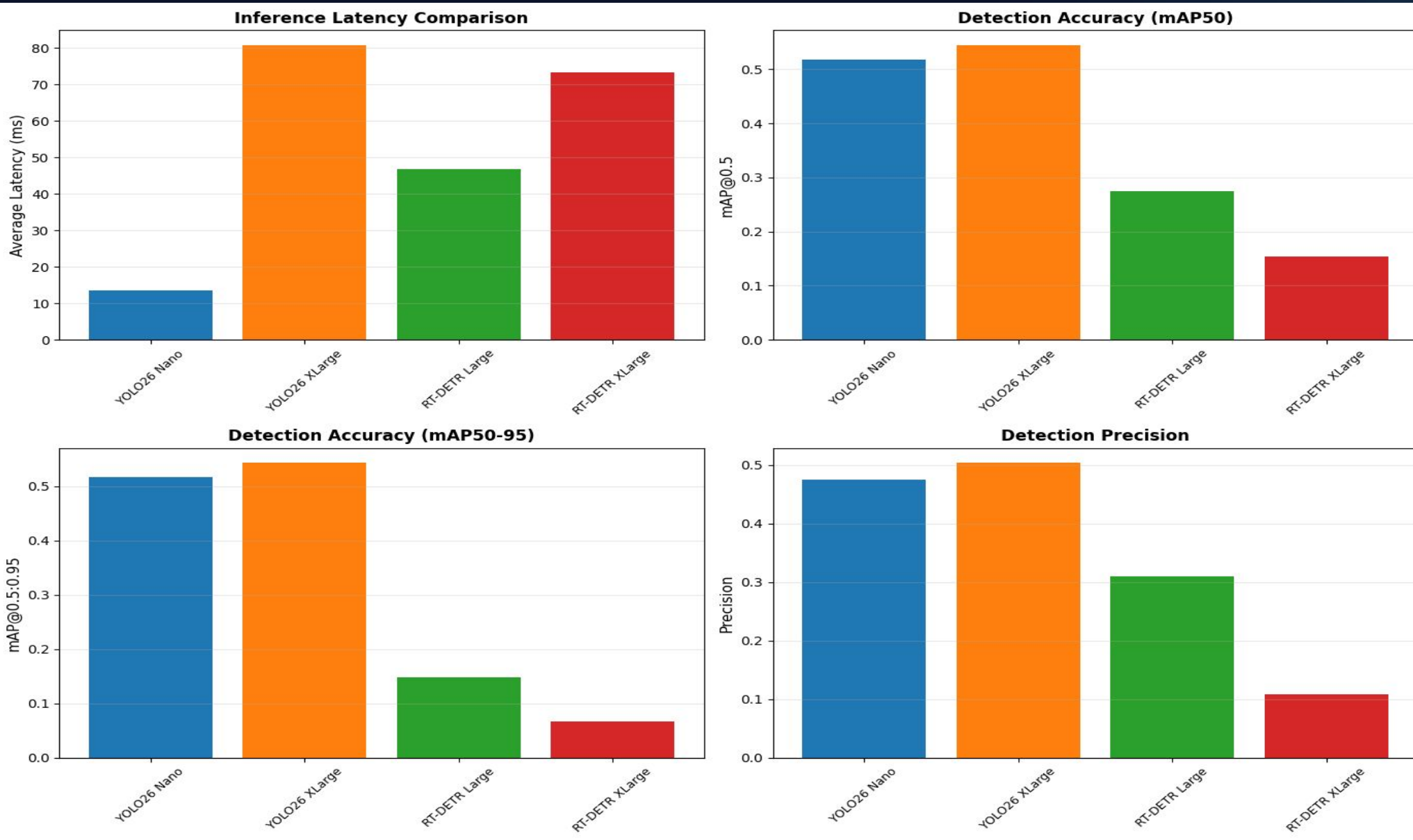


Iterative Training Execution: System automatically sweeps and trains 4 structural variants: YOLO26 Nano (batch 48), YOLO26 XLarge (batch 12), RT-DETR Large (batch 16), and RT-DETR XLarge (batch 8).



Standardized Optimization: Trained using AdamW optimizer for 100 epochs, early stopping patience of 20, and a fixed 640 x 640 tensor size to guarantee a fair benchmark across paradigms.

Project Demonstration Deliverables



Initial Validations: Latency vs Context Trade-off



Key Insight: YOLO26 yields significantly lower processing time per image, excelling in real-time edge constraints. Conversely, RT-DETR's global contextual reasoning yields lower false-negative rates for dense, overlapping pathologies.

Current Progress & Status

- ✓ Over 90% of the IEEE Draft is completed.
- ✓ 75%+ Code Implemented & Automated Pipeline functional.
- ✓ Core Hypothesis Validated (NMS elimination is viable).

Thank You